Statistics 280 – Honors Introduction to Statistics and Data Analysis

Fall 2026

MW 4 – 5:30 pm, 260 Weiser Hall

Instructor: Corey Powell

Email: cpow@umich.edu

Office: School of Kinesiology Building Room 4110

Office hours: F 1 - 2 pm, 3 - 4 pm or by appointment

Zoom: https://umich.zoom.us/j/6659109095

Graduate Student Instructors:

Devanshi Shah

Lab section: Th 2:30 - 4 pm, B760 East Hall

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Office hours: T 5:30 - 7 pm, F 10 - 11:30 am

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Zoom: https://umich.zoom.us/j/6032972250

Ali Fadel Alyaqoub

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Course information

Course webpage: https://umich.instructure.com/courses/862144

Waiting lists and overrides: Contact the undergraduate advisor Gina Cornaccia (ginalc@umich.edu) with questions about waiting lists and override codes.

Required Textbook: Statistics (4th edition). Authors: Freedman, Pisani, Purves

Required Software: We will be using the R programming language, which is free and open source. See https://www.r-project.org for more information. We will also be using RStudio as an interface to R. See https://techvidvan.com/tutorials/install-r/ for installing R and RStudio. We will help you install R and RStudio during the lab meetings this week.

A reference for most of the R that we will cover is available at https://rstudio-education.github.io/hopr/index.html We will not cover all of the material in this text, however, and will cover the material in a different order than the text. We will be covering roughly the following R topics:

- R as a calculator
- Data structures in R: vectors (lists), data frames (tables), matrices
- Functions
- If-then statements
- Loops
- Using vectors to replicate loops
- Basic plotting
- Simple linear regression and confidence intervals using *lm*
- Random sampling
- Basic simulation
- T-tests

Grading: I will base grades on the following, with a maximum possible total of 1,000 points:

- Quizzes: 300 points
- Class participation: 50 points
- Two in-class midterms: 200 points each

In-class final: 250 points.

Quizzes: I will assign homework every week, but we will not be grading the homework. There will also be lab exercises that we will not collect. We will post solutions to the homework once the due date has passed and lab exercises after the end of the lab. We will be using quizzes to test your understanding of the weekly homework and lab exercises. There will be 10 quizzes during the semester, with 5 quizzes during the last 20 minutes of lecture and 5 quizzes during the last 20 minutes of lab. The dates of the quizzes are below:

Lecture: 9/9, 9/23, 10/7, 10/28, 12/2

Lab: 9/17, 10/1, 10/22, 11/5, 11/19.

Each quiz will consist of 2-3 homework or lab exercises chosen from previous homeworks and labs. I will drop the lowest two quiz scores and scale the remaining 8 quizzes to be worth 300 points.

Class participation: There will be two components of class participation: attendance in lab sections and participation in office hours or lecture discussions. You will receive 5 points for each lab session that you attend, with a maximum of 40 points. You will also receive 5 points for each office hour visit, with a maximum of 10 points. If you do not come to office hours, then I will award up to 10 points based on your participation in lecture discussions.

Midterms: There will be in-class midterms on **Wednesday, October 14** and **Wednesday, November 11**. The midterms will be closed book, with no notes allowed. You will need to bring a calculator without internet access.

Final: The closed book, closed notes final exam will be in class on **Monday, December 14** from **4 - 6 pm**. The final will be cumulative, with an emphasis on the material covered since the second midterm. You will need to bring a calculator without internet access.

Missing midterms or the final: I will not give make-up exams. Please check for scheduling conflicts and mark your calendar now. If you miss

a midterm for an excusable reason (illness, family emergency, etc.), then the remaining midterm will count for 300 points and the final 350 points. The quizzes and class participation will count for 350 points. You will need to provide documentation of the excuse. You will need to withdraw from the course if you miss both midterms. You will receive an incomplete grade if you miss the final. The incomplete grade becomes a failing grade if you do not make up the final by the end of the 4th week of the next full term of enrollment. See https://lsa.umich.edu/lsa/academics/lsa-academic-policies-and-procedures/grading-and-transcript-notations/incompletes.html for more information.

Grading Scale

I will assign grades as follows:

Points	Grade	Points	Grade
950+	A+	650-699	C+
900-949	A	600-649	C
850-899	A-	550-599	C-
800-849	B+	500-549	D+
750-799	B	450-499	D
700-749	B-	400-449	D-

I may bump up borderline cases to the next higher grade based on your final exam score and improvement during the semester.

Study Pairs

We will randomly divide you into study pairs during the first lab session. We strongly encourage you to work on problem sets and lab exercises in these study pairs and/or larger study groups. The quizzes and exams must be your own work, however. Please see me if you haven't found a study group but would like to be part of one.

Course Policies

We will not tolerate academic dishonesty. We encourage working with others on homework assignments, but your exams must be your own work. Cheating will result in a 0 grade for the exam and could result in failing the class or more serious consequences. Exams will require the use of a scientific calculator that is capable of taking a square root. Calculators may not be shared, and cell phones and laptops may not be used. It is your responsibility to bring a calculator to the exam.

You should only use email for questions or concerns that relate to you personally, such as a missed class due to illness. You should post any other questions, such as questions about course material, clarification on course policies, etc., as discussion threads on Canvas, so that all students have access to the answer. I rarely check email in the evening or on weekends.

How to do well

Stay on top of the material. **This course moves quickly and is more theoretical than a typical introductory statistics course.** If you get behind, it will be difficult to catch up.

I post the lecture notes on Canvas before each lecture. Read the lecture notes and the corresponding textbook chapters before class.

Attend every lecture and lab. You are responsible for all material covered in lecture, and lecture will regularly cover material that is not in the textbook.

Ask questions! Ask questions in lecture, ask questions in lab, ask questions to members of your study group, ask questions in office hours, and ask questions in Canvas discussion threads. The most effective learning occurs when there is a dialogue between the student and the instructor. You will want to ask a question as soon as you don't understand something. It is likely that you will not understand later material if you do not resolve questions quickly. It is fine to ask "I did not understand that. Could you please explain that again?"

You may struggle with the programming component of the course, especially if you have not programmed before. Please ask lots of questions in lab, come to our office hours, create discussion threads on Canvas, and work with members of your study group. It is important to resolve questions quickly so that they do not become major obstacles later on. We are here to help.

Work practice problems. The book is an excellent source of practice problems, and lectures and labs give additional practice exercises. At least a few questions on every test will be closely related to either: a) the book exercises, b) the lecture review exercises, or c) the lab exercises. Work as many of these problems as you can.

Course schedule

Part 1: Study design, descriptive statistics, correlation, and regression

Lecture 1: Course introduction, thinking statistically, experiments (Ch. 1)

Lecture 2: Observational studies, natural experiments (Ch. 2)

September 7: Labor Day

Lecture 3: Univariate descriptive statistics (Ch. 3-4), Quiz 1

Lecture 4: The normal distribution, Markov and Chebyshev inequalities (Ch. 5)

Lecture 5: Correlation (Ch. 8-9), Quiz 2 in lab Thursday

Lecture 6: Simple linear regression (Ch. 10-12)

Lecture 7: Regression, continued, Quiz 3

Part 2: Probability and sampling

Lecture 8: Introduction to probability (Ch. 13-14)

Lecture 9: Bayes' rule, Quiz 4 in lab Thursday

Lecture 10: The binomial and hypergeometric distributions (Ch. 15)

Lecture 11: Random variables, expected value, standard error, Quiz 5

Lecture 12: Box models, repeated sampling (Ch. 16-17)

Midterm 1 (Covers Part 1)

Fall Break October 19-20

Lecture 13: The Central Limit Theorem (Ch. 18)

Lecture 14: Sample surveys (Ch. 19-20), Quiz 6 in lab Thursday

Part 3: Inference

Lecture 15: Confidence intervals I (Ch. 21, 23), Quiz 7

Lecture 16: Confidence intervals II, the Gauss model (Ch. 6, 24)

Lecture 17: Hypothesis testing I (Ch. 26), Quiz 8 in lab Thursday

Lecture 18: Hypothesis testing II (Ch. 27)

Midterm 2 (Covers Part 2)

Lecture 19: Hypothesis testing III (Ch. 29)

Lecture 20: Hypothesis testing IV, Quiz 9 in lab Thursday

Part 4: Chi-squared distribution, connection between confidence intervals and p-values

Lecture 21: The Chi-squared distribution

Thanksgiving Break: November 25 - 27

Lecture 22: More on the Chi-squared distribution, goodness of fit tests (Chapter 28)

Lecture 23: Goodness of fit tests, tests of independence (Chapter 28), Quiz 10

Lecture 24: Connection between confidence intervals and p-values, TBA

Review: Lecture 25

Final: Monday December 14 from 4-6 pm in Weiser Hall Room 260